



NAKKA HEMANTH DATTA

Engineering Physics | EP23B028

CGPA: 8.45 | GitHub | LinkedIn | Kaggle

Education

- Indian Institute of Technology Madras** Chennai, TN
B. Tech – Major: Engg. Physics, Minor: AIML Jul '23 – Jul '27
- JNV, Pondicherry** Pondicherry, PY
Class XII (CBSE) | 95.80% 2023

Professional Experience

- Wells Fargo | PPO Offered** Bangalore, India
Tech Developer Intern | AI Systems & LLMOps May '26 – Jul '26
- Developed & benchmarked **LLM-as-a-Judge evaluation pipelines** for hallucination, truthfulness, & response quality, using **judge-vs-gold calibration** to quantify evaluator reliability.
- Built **trace-based observability** across multi-step AI agent execution, analyzing **latency, token consumption, & model input/output behavior** to diagnose performance & behavioral degradation.
- Architected a **PDF document extraction pipeline** with PyMuPDF, automating document parsing, **JSON-to-SQL schema normalization**, & structured-data reconciliation.
- Built a **containerized FastAPI backend** for enterprise workflows, integrating structured data processing with production APIs & scalable service components.

- CV. OMSAI INFORMASI TEKNOLOGI** Remote, Singapore
AI/ML Engineer (Contract) Feb '26 – May '26
- Designed a **multimodal Corrective RAG (CRAG) pipeline** combining dense embeddings, BM25 retrieval, query rewriting, & reranking with **LangGraph & ChromaDB**.
- Developed a **table-aware retrieval architecture** using document-level summaries, row-level embeddings, & raw-table injection, achieving **100% table-query accuracy** across **250+ enterprise documents**.
- Applied **semantic chunking & Ragas-based context precision/recall evaluation** to identify & prune irrelevant retrievals, improving answer groundedness while reducing LLM inference latency.
- Reduced **prompt token consumption by 35%** through retrieval-quality optimization & context filtering, while building citation-level evaluation & visualization for retrieval debugging.

Technical Skills

- Languages & Data:** Python, C/C++, SQL, NumPy, pandas, MATLAB, Mathematica
- Machine Learning:** PyTorch, TensorFlow, scikit-learn, Transformers, XGBoost, CatBoost, LightGBM, Optuna, OpenCV
- ML Methods:** Classification, Feature Engineering, Hyperparameter Optimization, Parameter-Efficient Fine-Tuning, Generative Modeling
- AI / Retrieval:** RAG/CRAG, FAISS, ChromaDB, LangGraph, PEFT (LoRA/BitFit), Ragas, LLM-as-a-Judge
- Tools:** FastAPI, Docker, Kubernetes, Redis, Git

Relevant Coursework

- Intro to Machine Learning
- Introduction to Deep Learning
- Natural Language Processing
- Computer Vision
- Probability & Statistics
- Data Structures & Algorithms
- Database Management Systems
- Computational Physics
- Digital Signal Processing
- Linear Algebra

Hackathons

- 2nd Prize, Mando AI Tool Dev Hackathon 2025** May '25 – Jun '25
LangGraph | Claude LLM | FAISS | Azure
- Engineered a full-stack document QA system supporting PDF, DOCX, & CSV, using LangGraph, VoyageAI embeddings, Claude LLM, & FAISS hybrid search across **2,000+ page documents**.

- Round 3 Finalist, Appian Algnite Hackathon 2025** Mar '25 – Apr '25
CLIP | Gemini API | FAISS | Azure
- Built a multimodal product discovery platform using CLIP+FAISS hybrid search across **14K+ products**, with a Gemini-powered AI browser extension for automated e-commerce checkout.

Mentorship

- Avanti Fellows & EE Intern Buddy Program**
Student Mentor Sep '23 – Aug '26
- Mentored underprivileged JEE aspirants by giving back as an Avanti Fellows alumnus & guided junior EE students as Mentor.

Research & Patents

Gesturize: Edge Sign Language Translation System

Indian Patent: 202541022908 | CFI, IIT Madras Apr '24 – Mar '25

- Developed a real-time sign-language recognition pipeline for **Raspberry Pi**, transitioning from attention-based seq2seq to a **few-shot Siamese network** for low-resource inference.
- Extracted **MediaPipe hand-landmark features** with interpolation & synthetic gesture augmentation, & built an **English-to-ISL grammar transformation pipeline** (Stanford Parser, spaCy) to generate **synthetic training data**.
- Led annotation of a **60+ volunteer-collected open-source ISL dataset** at CFI, establishing a clean training corpus for downstream recognition models.

Projects

Foundational Model Adaptation Pipeline

Under Prof. Sukhendu Das | CS6350, IIT Madras Jul '25 – Dec '25

- Evaluated **CLIP & DINOv2** for fine-grained classification across **zero-shot, few-shot, & PEFT** regimes.
- Implemented **LoRA, BitFit, Prefix Tuning**, updating only **0.7% of parameters** achieving **94.67% classification accuracy** with DINOv2.
- Designed a **two-stage adaptation strategy** separating backbone feature extraction from classifier optimization for efficient model experimentation & iteration.

Score-Based Generative Model Evaluation Framework

Under Prof. Sheetal Kalyani | EE5180, IIT Madras Jul '25 – Dec '25

- Implemented **DDPM, DDPM++, NCSN++** image-generation pipelines for CIFAR-10 across **VP, VE, & Sub-VP SDE** formulations in PyTorch.
- Conducted controlled ablations across **12 architectures & 200K training steps**, evaluating generated image quality using **FID & Inception Score**.
- Achieved **FID 10.52 / IS 7.37** with VP-DDPM++ and **FID 11.34 / IS 7.03** with VE-NCSN++, identifying **width-over-depth** as a key architectural trade-off.

Customer Analytics & Churn Prediction System

Python | XGBoost | CatBoost | Optuna Jan '25 – May '25

- Developed an end-to-end **customer churn prediction pipeline**, benchmarking **Logistic Regression, XGBoost, CatBoost, & LightGBM** to achieve **86% accuracy & 0.84 F1-score**.
- Engineered **60+ domain features** & performed **100+ Optuna hyperparameter trials**, improving model performance by **17%** & reducing **false positives by 23%**.

Time Series Forecasting – Data Size Impact Study

Python | Scikit-learn | TensorFlow | KNN | SVR | LSTM Jan '25 – May '25

- Compared **KNN, SVR, & LSTM** for **AAPL stock-price forecasting** across **100–1,174 training samples**; found **LSTM dominance beyond ~1,000 samples**, reaching **0.00042 MSE / 0.01603 MAE** versus KNN's 0.00052 / 0.01768.
- Engineered a **leakage-aware time-series pipeline** with splitting **MinMax scaling, 10-day sliding windows** for next-day price prediction.

Multimodal Vision-Language Adaptor (“Minivision”)

PyTorch | Transformers | 250M-parameter model Jan '26 – May '26

- Developed a **vision-language adaptor** aligning image embeddings with a text LLM, training a **250M-parameter multimodal model** on **25K+ curated image-text pairs**, with a checkpoint benchmarking framework for **real-time visual QA evaluation**.

Distributed High-Performance N-Body Simulation Engine

Under Prof. Yasir Iqbal | PH5730, IIT Madras Jan '26 – May '26

- Built a scalable N-body simulation pipeline using **NumPy vectorization, Barnes-Hut, multiprocessing, & distributed execution**, achieving **~78x speedup** over the naive baseline.
- Parallelized Barnes-Hut traversal across **8 workers** for an additional **~3x speedup** & implemented checkpointed coordinator-worker execution with **< 0.1% overhead**.

Scholastic Achievements

- CBSE Certificate of Merit for **top 0.1% rank (100/100) in CS**, Class XII Boards 2023 & Reliance UG Scholarship.
- AIR 5476** in JEE Advanced 2023 (2.5L+ candidates) & **Top 1.42%** in JEE Mains 2023.

Extra-Curriculars

- NSO Volleyball & state/national chess player & EEA frisbee champions